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Enhancing Handwritten Text Recognition Using Transformer Models Derived from Large Language Models

(تعزيز التعرف علي النصوص اليدوية باستخدام نماذج المحولات المستمدة من
نماذج لغوية كبيرة)

A Thesis Submitted in Partial Fulfillment of the Requirements for

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Chapter One

Introduction

1. CHAPTER I

1.1. Introduction:

In the vast realm of artificial intelligence, computer vision emerges as a beacon of innovation, striving to imbue machines with the ability to comprehend and interpret the visual intricacies of our world. With a primary focus on decoding visual data, such as images and videos, this field entails the development of sophisticated algorithms and techniques. These advancements empower computers to recognize objects, discern patterns, and contextualize the information they perceive. Amid the challenges faced in the realm of character recognition, deciphering handwritten characters stands out as a particularly intricate task. The diversity in writing styles further complicates the function of character recognition. Recent strides in this domain have been made possible by the remarkable progress of Convolutional Neural Networks (CNNs). Their prowess extends to image-based recognition, video analytics, and natural language processing, ushering in a new era of possibilities (Skala, V., Singh, T.P., Choudhury, T., Tomar, R. and Bashar, M.A. eds., Md. Abul Bashar Editors, 2022).

A pivotal application of computer vision lies in the preservation of cultural heritage through the recognition of handwritten text on historical documents. Recognizing the significance of this task, the last decade has witnessed a shift towards the adoption of Recurrent Neural Networks (RNNs) coupled with Connectionist Temporal Classification (CTC). This combination forms an end-to-end system capable of transcribing raw images containing entire lines of text. In recent Handwritten Text Recognition (HTR) contests, RNN-CTC methods have exhibited the lowest error rates, with multidimensional Long Short-Term Memory (MDLSTM) networks at their core, offering recurrent connections along each dimension in the data (Sahare, et al., 2018).

Notably, the importance of HTR transcends the boundaries of historical document preservation, finding applications in document digitization and even the healthcare sector. The interpretation of handwritten medical prescriptions becomes a critical task, where traditional approaches relying on RNNs or CNNs have showcased significant progress. However, the landscape is evolving with the advent of Large Language Models (LLMs) and transformer architectures. These models, renowned for their exceptional performance in natural language processing, present an exciting opportunity to revolutionize HTR and redefine the boundaries of what is achievable in this dynamic field.

1.2. Motivation:

Recent advances in deep learning and transformer-based approaches offer opportunities to improve the accuracy and efficiency of handwritten text recognition (HTR). By exploring the potential of these approaches to improve HTR, this research aims to support the development of more effective recognition tools for HTR, potentially revolutionizing the way the handwritten text is recognized.

Furthermore, this research is motivated by growing interest in the application of artificial intelligence (AI) in optical character recognition (OCR) technologies.

By investigating the use of these approaches in handwritten text recognition (HTR), this study aims to contribute to the growing literature on the application of AI in computer vision field and pave the way for more innovative and effective OCR tools in the future.

1.3. Problem Statement:

This research seeks to address the inherent challenges within Handwritten Text Recognition (HTR), particularly focusing on the limitations associated with current methodologies. The recognition of handwritten characters poses significant difficulties due to variations in size, shape, and writing styles among individuals. Traditional approaches, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), while demonstrating progress, exhibit limitations in handling the intricacies of diverse handwriting styles.

1.4. Aims and Objectives:

1.4.1. Aims:

The aim of this research is to enhance handwritten text recognition by leveraging transformer models derived from large language models.

1.4.2. Objectives:

The specific objectives of this research are:

1. Literature Review and Methodology Investigation:

Conduct a comprehensive review of existing HTR methods, with a focus on traditional approaches and their limitations.

2. Adaptation of Transformer Architecture:

Refine transformer architectures for handwritten text recognition challenges by customizing attention mechanisms, positional encoding, and self-attention mechanisms to suit the unique spatial characteristics of handwriting.

3. Data Collection and Model Training:

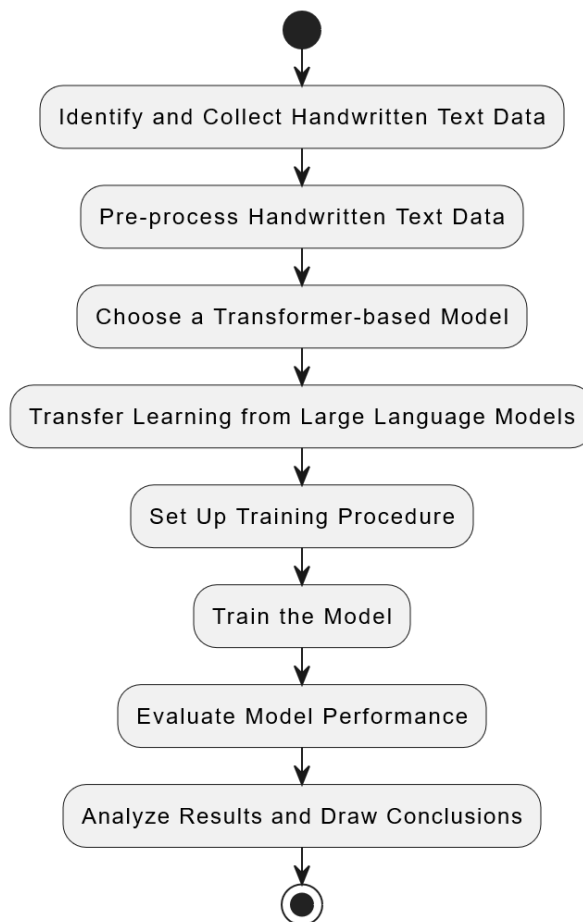
Assemble a diverse dataset of handwritten text samples, emphasizing variability in handwriting styles. Give specific attention to practical applications, such as medical prescriptions. Train transformer-based models on the compiled dataset, employing transfer learning from pre-trained Large Language Models (LLMs), and implement fine-tuning strategies to improve the model's capability in recognizing handwritten text.

4. Performance Assessment and Generalization:

Evaluate the performance of the proposed models using standard metrics like accuracy, precision, recall, and F1 score. Conduct comparative analyses with baseline models and traditional HTR approaches. Investigate the robustness and generalization capabilities of the proposed models across diverse handwriting styles.

1.5. Research Methodology:

This methodology outlines a comprehensive approach to advancing handwritten text recognition through the strategic integration of the IAM Handwriting Dataset, Transformer-based models, and transfer learning from Large Language Models. By meticulously addressing data selection, preprocessing, model training, and performance evaluation, this research methodology aims to enhance the understanding and application of sophisticated language models in the specialized context of handwritten text recognition.



1. Identify and Collect Handwritten Text Data

Selecting the "IAM Handwriting Dataset" is a judicious choice in alignment with the objectives of my handwritten text recognition research. I will ensure that this dataset incorporates a diverse range of handwriting samples, thereby fostering enhanced generalization capabilities within the model.

2. Pre-process Handwritten Text Data

Implement pre-processing steps such as normalization, data augmentation, and resizing to prepare the data for training.

3. Choose a Transformer-based Model

Selecting an open-source Large Language Model (LLM) such as GPT-3, BERT, GPT-2, XLNet, or a similar architecture is a strategic decision for my handwritten text recognition research. The next step involves determining the most suitable transformer model architecture among these options. Additionally, careful consideration will be given to any necessary modifications or the incorporation of additional layers tailored to the specific requirements of the handwritten text recognition task.

4. Transfer Learning from Large Language Models

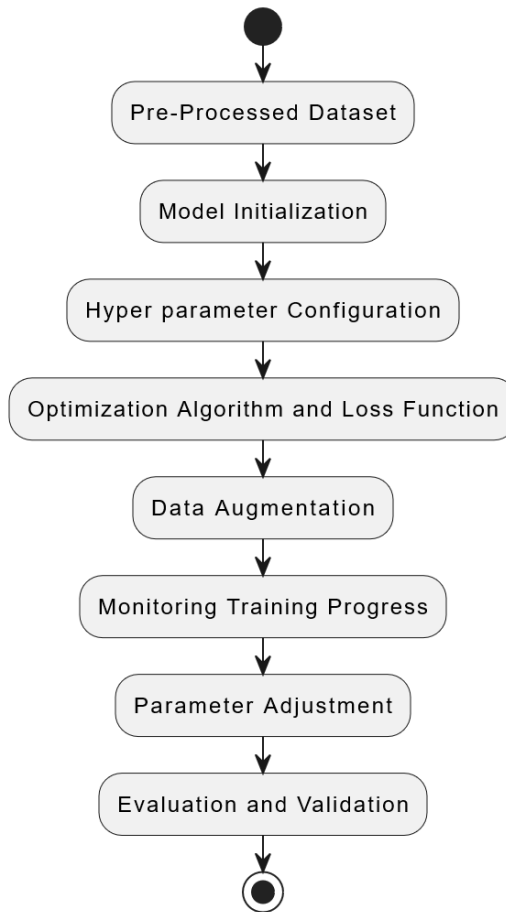
Leveraging transfer learning from large language models (LLMs) is a pivotal strategy in the context of my research on handwritten text recognition. This approach involves harnessing the pre-existing knowledge encoded within sophisticated language models such as GPT-3, BERT, GPT-2, XLNet, or similar architectures and adapting this knowledge to enhance the performance of my model on the specific task of recognizing handwritten text.

5. Set Up Training Procedure

Establishing the training procedure is a critical step in the development of a handwritten text recognition model. This involves defining various components, including hyper parameters, optimization algorithms, convergence criteria, and optionally, exploring parallel or distributed training methods to efficiently utilize available resources.

6. Train the Model

The training phase involves the implementation of the proposed transformer model on the pre-processed handwritten text dataset. The following steps outline the process:



a. **Pre-Processed Dataset**

Utilize the selected handwritten text dataset, which has undergone pre-processing steps to ensure uniformity, cleanliness, and compatibility with the transformer model.

b. **Model Initialization**

Initialize the transformer model architecture based on the chosen large language model (LLM), incorporating any modifications or additional layers necessary for handwritten text recognition.

c. **Hyper parameter Configuration**

Set hyper parameters such as learning rate, batch size, and the number of training epochs. These values should be determined based on experimentation and adjusted to optimize the convergence and accuracy of the model.

d. **Optimization Algorithm and Loss Function**

Implement the chosen optimization algorithm, such as Adam, along with an appropriate loss function tailored to the task of handwritten text recognition. Ensure that the loss function aligns with the objectives of the research.

e. **Data Augmentation**

Apply data augmentation techniques during training to introduce variability into the dataset, enhancing the model's ability to generalize to different handwriting styles.

f. **Monitoring Training Progress**

Implement logging and monitoring mechanisms to track key metrics, including training and validation loss, accuracy, and any other relevant performance indicators. Monitor convergence trends to determine when the model is sufficiently trained.

g. **Parameter Adjustment**

Regularly evaluate the model's performance on validation data and be prepared to adjust hyper parameters or other configurations as needed. This iterative process ensures continual refinement of the model throughout the training phase.

h. **Evaluation and Validation**

Periodically assess the model's performance on a separate validation set to ensure that it generalizes well to unseen data. Fine-tune parameters based on the observed results.

By systematically executing these steps, the training phase aims to empower the transformer model with the ability to effectively recognize handwritten text, leveraging insights derived from large language models for enhanced performance in the specific context of the research objectives.

7. Evaluate Model Performance

Define metrics for evaluating model performance, such as accuracy, precision, recall, and F1 score. Compare the results with baseline models or existing methods for benchmarking.

8. Analyze Results and Draw Conclusions

In the final stage of the research process, the analysis of results and drawing conclusions is pivotal to understanding the effectiveness of the proposed approach in enhancing handwritten text recognition using transformer models derived from large language models. The following steps outline how this phase can be conducted:

a) **Quantitative Results Presentation:**

Present the quantitative results obtained from the experiments, including metrics such as accuracy, precision, recall, F1 score, and any other relevant evaluation criteria. Use visualizations such as charts or graphs to highlight trends and patterns in the performance of the model over different training epochs.

b) **Qualitative Results Presentation:**

Provide qualitative insights by showcasing examples of handwritten text recognition outputs generated by the model. This can include side-by-side comparisons of the model's predictions against ground truth labels, emphasizing instances where the model excels or faces challenges.

c) **Comparison with Baseline Models:**

Compare the performance of the proposed transformer model with baseline models or traditional handwritten text recognition approaches. Highlight improvements or differences in accuracy and efficiency achieved through the novel approach.

1.6. Research Scope:

The research scope centers on optimizing Handwritten Text Recognition (HTR) specifically for English language handwritten text, leveraging a pre-trained model trained on the "IAM Handwriting" dataset its contains around 1,539 text pages. Each page typically includes multiple lines of handwritten text, contributing to a diverse set of samples for training and evaluation purposes.

1.7. Research Planning:

- Weeks 1-4: Literature Review and Model Design
- Weeks 5-7: Dataset Collection and Preprocessing
- Weeks 8-10: Model Training and Fine-Tuning
- Weeks 11-12: Evaluation and Analysis, Thesis Writing

1.8. The Thesis Organization

This research organized as following:

Chapter I: contains the research problem statement and objectives.

Chapter II: discusses the literature review and related work.

Chapter III: describes the research methodology and the implementation of the techniques used

Chapter IV: presents the Experiments and results. Finally.

Chapter V: concludes this research and presents Recommendations for future works.

Chapter Two

Literature Review

2.CHAPTER II

2.1. Introduction

Recent advancements in handwritten text recognition (HTR) have witnessed a paradigm shift, driven by the integration of deep learning methodologies. Notably, the convergence of transformer-based models, pre-trained architectures, and innovative training strategies has propelled HTR into new realms of accuracy and efficiency. This literature review explores diverse approaches, ranging from fine-tuning vision encoder-decoder transformers for historical document transcription to the development of novel architectures like TrOCR and Handwriting Transformers. The synthesis of these studies not only underscores the transformative potential of deep learning in HTR but also highlights the ongoing evolution towards more effective, efficient, and versatile recognition systems.

2.2. Problem Background

In the vast realm of artificial intelligence, the evolution of computer vision emerges as a captivating journey, a quest to empower machines with the extraordinary ability to decipher the intricate language of visuals. At the heart of this narrative lies a profound challenge – the recognition of handwritten characters, a task made all the more complex by the kaleidoscope of diverse writing styles. In this unfolding saga, Convolutional Neural Networks (CNNs) have played a pivotal role, advancing the frontier of image-based recognition. Yet, recent chapters in the story of Handwritten Text Recognition (HTR) reveal a shift towards a new protagonist – Recurrent Neural Networks (RNNs) coupled with Connectionist Temporal Classification (CTC), orchestrating the transcription of entire lines of text.

Beyond the realms of character recognition, our tale weaves through the rich tapestry of cultural heritage preservation, historical document analysis, and the digitization of documents, each page holding a story waiting to be told. In the corridors of healthcare, the narrative takes a critical turn as handwritten medical prescriptions demand interpretation. However, the journey encounters challenges – the limitations of traditional approaches relying on RNNs and CNNs to navigate the diverse landscape of handwriting styles.

Cue the turning point – a surge in deep learning and the advent of transformer-based approaches. Our story is now infused with the promise of improvement, as we seize the opportunities presented by Large Language Models (LLMs) and transformer architectures, celebrated champions in the arena of natural language processing. The stage is set for a redefinition of boundaries in the dynamic field of HTR.

The plot thickens as we confront the problem statement – an exploration of the inherent challenges in HTR, a focused gaze on the limitations of existing methodologies in decoding the unique variations in size, shape, and styles of handwritten characters. The traditional actors, RNNs and CNNs, have made strides but struggle with the nuances of diverse handwriting styles.

Our hero emerges with a clear aim – to enhance handwritten text recognition through the strategic leverage of transformer models born from large language models. The narrative unfolds with objectives that read like chapters – a comprehensive literature review to set the stage, the adaptation of transformer architecture to confront HTR challenges head-on, a quest for diverse data and rigorous model training, with a spotlight on practical applications like medical prescriptions. The climax arrives with a performance assessment, an evaluation of our hero's journey across diverse handwriting styles, a testament to its versatility.

In essence, our story is an epic, a journey that seeks to bridge the gap between the familiar terrain of traditional HTR methods and the uncharted territories of the transformative potential offered by transformer models. It is a story of addressing nuanced challenges, of unlocking the secrets encoded in handwritten characters, and ultimately, a tale of paving the way for more effective and versatile HTR tools in the ever-evolving saga of artificial intelligence.

2.3. Images Processing

Embarking on the realm of pixels and patterns, image processing emerges as a transformative art. This method, a digital alchemy, orchestrates operations to breathe life into images, birthing enhanced versions or revealing hidden insights. In the technological symphony, image processing stands tall among swiftly evolving disciplines. It weaves its

magic as a core research domain, entwining engineering and computer science in its tapestry of possibilities.

Delving deeper, image processing is the digital metamorphosis of visuals, a conversion of images into a binary language that unveils their secrets. This method, a signal dance, transforms pixels into a canvas where operations unfold, crafting a narrative of enhanced visuals or extracted intelligence. As (Covington, Michael A, 2009) echoes, image processing is the art of translating pixels into a digital language, a symphony where inputs are images, and outputs are reborn images or their inherent characteristics. It's a dance of signals, a harmonious interplay between what the eye sees and what the mind deciphers. In essence, image processing is more than a method – it's a voyage into the heart of visuals, unraveling their mysteries and enhancing their tales through the wizardry of algorithms and signals (Naveenkumar, , M. and V. Ayyasamy., 2016).

2.4. Optical Character Recognition (OCR)

Optical character recognition refers to the branch of computer science that involves reading text from paper and translating the images into a form that the computer can manipulate (for example, into ASCII codes).

An OCR system enables you to take a book or a magazine article, feed it directly into an electronic computer file, and then edit the file using a word processor. All OCR systems include an optical scanner for reading text, and sophisticated software for analyzing images. Most OCR systems use a combination of hardware (specialized circuit boards) and software to recognize characters, although some inexpensive systems do it entirely through software. Advanced OCR systems can read text in large variety of fonts, but they still have difficulty with handwritten text. It is the mechanical or electronic translation of scanned images of handwritten, typewritten or printed text into machine-encoded text. It is widely used to convert books and documents into electronic files, to computerize a record-keeping system in an office, or to publish the text on a website.

OCR makes it possible to edit the text, search for a word or phrase, store it more compactly, display or print a copy free of scanning artifacts, and apply techniques such as machine translation, text-to-speech and text mining to it. OCR is a field of research in pattern recognition, artificial intelligence and computer vision.

OCR systems require calibration to read a specific font; early versions needed to be programmed with images of each character, and worked on one font at a time. "Intelligent" systems with a high degree of recognition accuracy for most fonts are now common. Some systems are capable of reproducing formatted output that closely approximates the original scanned page including images, columns and other non-textual components.

Optical character recognition (OCR) is one of the most popular areas of research in pattern recognition because of its immense application potential. However, most of the available methods deal with the recognition of the Roman script and some of the oriental scripts like Kanji, Kana, etc. The purpose of this OCR system is to take handwritten English characters as input, process the character, train the neural network algorithm, to recognize the pattern and modify the character to a beautified version of the input. This work is restricted to English characters and numerals only. It is also helpful in recognizing special characters. It can be further developed to recognize the characters of different languages.

One of the primary means by which computers are endowed with humanlike abilities is through the use of a neural network. Neural networks are particularly useful for solving problems that cannot be expressed as a series of steps, such as recognizing patterns, classifying them into groups, series prediction and data mining (Isheawy, N.A.M. and Hasan, H., 2015).

2.5. Handwritten Text Recognition.

In the expansive domain of artificial intelligence and computer vision, the scientific exploration of Handwritten Text Recognition (HTR) unfurls as a multifaceted journey, intricately woven with challenges and technological evolution.

The scientific inquiry commences with the underpinning of computer vision, a discipline committed to imparting machines with the perceptual acuity to comprehend and interpret visual data. At the forefront of this exploration lies the nuanced task of recognizing handwritten characters, encapsulating the diverse spectrum of human writing styles.

Convolutional Neural Networks (CNNs) emerge as stalwarts, advancing image-based recognition and laying the foundation for decoding visual data. Recent forays into HTR navigate the complexities of character recognition using Recurrent Neural Networks (RNNs) coupled with Connectionist Temporal Classification (CTC), particularly adept at transcribing entire lines of handwritten text (Sahare, et al., 2018)

Beyond the realm of mere character recognition, the significance of HTR extends into the preservation of cultural heritage. Recurrent connections within multidimensional Long Short-Term Memory (MDLSTM) networks unveil a transformative potential, offering a digital refuge for historical documents (Skala, V., Singh, T.P., Choudhury, T., Tomar, R. and Bashar, M.A. eds., Md. Abul Bashar Editors, 2022).

Motivated by the surge in deep learning, the scientific landscape expands to embrace transformer architectures. Large Language Models (LLMs) and transformers, lauded for their prowess in natural language processing, beckon HTR into a new era of possibilities. This scientific trajectory seeks to bridge the chasm between traditional methodologies and the transformative capacities of transformers, redefining the theoretical boundaries within the dynamic field of HTR.

Amidst this scientific odyssey, the problem statement crystallizes around the intricacies of handwritten characters — variations in size, shape, and style pose formidable challenges. Traditional approaches like RNNs and CNNs, while demonstrating efficacy, grapple with the subtleties inherent in diverse handwriting styles.

The overarching scientific aim materializes — to augment handwritten text recognition by leveraging transformer models derived from large language models. The scientific objectives unfold meticulously: a comprehensive survey of existing HTR methods, the nuanced adaptation of transformer architecture to accommodate the intricacies of handwriting, meticulous data curation, and model training with a spotlight on practical applications, and a thorough scientific evaluation, dissecting robustness and generalization across diverse handwriting styles.

In this scientific narrative, HTR transcends its technological façade. It becomes a scientific guardian of cultural artifacts, a tool in the healthcare arsenal, and a testament to the evolving synergy between artificial intelligence and the intricacies of human

expression. This is the scientific overview of Handwritten Text Recognition — a landscape where algorithms grapple with the complexities of handwritten scripts, pushing the boundaries of scientific understanding and technological application.

2.6. Related Handwritten Text Recognition (HTR) Researches

The field of Handwritten Text Recognition (HTR) has witnessed notable advancements in recent years, spurred by the integration of deep learning techniques and the exploration of transformer-based models. However, several challenges persist, motivating ongoing research endeavors aimed at addressing key issues in HTR.

Historical documents pose a unique set of challenges for HTR due to their often degraded and diverse nature. Research by (Parres, D. and Paredes, R., 2023, August) identifies the transcription of historical documents as a particularly critical area within HTR. The existing dominance of convolutional recurrent neural networks (CRNN) with connectionist temporal classification loss in historical document recognition underscores the need for novel approaches. Despite the transformative impact of transformer models in natural language processing, their adoption in the context of historical document HTR remains limited.

TrOCR, a novel approach proposed by (Li, M., Lv, T., Chen, J., Cui, L., Lu, Y., Florencio, D., Zhang, C., Li, Z. and Wei, F., 2023, June.), addresses the persisting challenges in text recognition for document digitalization. Traditional methods relying on convolutional neural networks (CNNs) and recurrent neural networks (RNNs) for image understanding and text generation, respectively, often require additional language models for post-processing. TrOCR introduces an end-to-end solution with pre-trained image and text transformers, showcasing superior performance across printed, handwritten, and scene text recognition tasks. This indicates a promising avenue for achieving comprehensive text recognition without the need for additional post-processing steps.

The significance of transforming handwritten text images is further emphasized by (Bhunja, A.K., Khan, S., Cholakal, H., Anwer, R.M., Khan, F.S. and Shah, M., 2021), who propose Handwriting Transformers (HWT). HWT leverages a transformer-based network for styled handwritten text image generation, capturing both style-content entanglement and global/local style patterns. This innovative approach demonstrates the

potential to generate realistic styled handwritten text images, showcasing versatility in handling various writing styles and lengths of text.

Attention-based models, such as AttentionHTR proposed by (Kass, D. and Vats, E., 2022), tackle the challenge of limited training data in handwritten word recognition. By leveraging pre-trained models on scene text images and introducing a bidirectional LSTM-based sequence model with content-based attention, AttentionHTR proves effective on datasets like Imgur5K and IAM. This emphasizes the significance of transfer learning from synthetic scene text images to overcome data scarcity and enhance recognition performance.

Recognizing the need for effective word and character-level classification in handwritten documents (Balci, B., Saadati, D. and Shiferaw, D., 2017) and (Manchala, S.Y., Kinthali, J., Kotha, K., Kumar, J.J.K.S. and Jayalaxmi, J., 2020) focus on leveraging deep learning techniques with the IAM Handwriting Dataset. Both studies explore different architectures, including CNNs, LSTMs, and variants like VGG-19 and RESNET-18/34. These approaches demonstrate successful training for word-level classification, character segmentation, and achieving accuracy rates exceeding 90.3%.

The introduction of a Light Transformer-Based Architecture by (Barrere, K., Soullard, Y., Lemaitre, A. and Coüasnon, B., , 2022) addresses the limitations of traditional Transformer models in HTR, particularly when confronted with limited annotated data. This approach incorporates convolutional layers and Transformer components, maintaining reduced parameter counts suitable for small datasets. The hybrid loss function, combining Connectionist Temporal Classification (CTC) and Cross-Entropy (CE), contributes to training efficiency and improved performance, with the model achieving a Character Error Rate (CER) of 5.70% on the IAM test set.

HTR-Flor, presented by (de Sousa Neto, A.F., Bezerra, B.L.D., Toselli, A.H. and Lima, E.B., 2020), introduces a Gated Convolutional Neural Network with Bidirectional Gated Recurrent Units (Gated-CNN-BGRU) to address challenges in recognizing offline handwritten text. This model outperforms state-of-the-art CRNN approaches with fewer parameters, showcasing its efficiency in handling long sentences, variations, and noise in scanned documents. The success of HTR-Flor further emphasizes the potential for compact architectures in achieving superior recognition performance.

The study by (Coquenet, D., Chatelain, C. and Paquet, T., , 2020) introduces a Gated Fully Convolutional Network (GFCN) as a novel approach for unconstrained handwritten text recognition. By eliminating the need for Long Short-Term Memory (LSTM) cells and adopting convolutional layers with a gating mechanism, GFCN achieves competitive performance with fewer parameters. The incorporation of Depthwise Separable Convolutions (DSC) and a Fully Convolutional Network (FCN) further contributes to effective dependency modeling, showcasing the potential for innovative architectures in unconstrained HTR tasks.

In conclusion, while recent research has made significant strides in improving HTR through deep learning and transformer-based models, challenges remain in adapting these advancements to historical documents, addressing data scarcity, and optimizing model architectures for efficient recognition across various types of handwritten text. The collective insights from these studies lay the groundwork for continued exploration and innovation in the field of Handwritten Text Recognition.

2.7. Related HTR Researches Summary Table:

#	Study	Objective	Dataset	Tools & Techniques	Results	Opinion
1	Fine-Tuning Vision Encoder–Decoder Transformers for Handwriting Text Recognition on Historical Documents	Develop and assess a novel approach for Handwritten Text Recognition (HTR) on historical documents by fine-tuning pre-trained transformer models, specifically vision encoder–decoder models, with a focus on addressing challenges like	ICFHR 2014 Bentham dataset, ICFHR 2016 Ratsprotokolle dataset, Saint Gall dataset	Fine-tuning pre-trained transformer, particularly vision encoder–decoder models.	ICFHR 2014 Bentham dataset: 6.9% WER, ICFHR 2016 Ratsprotokolle dataset: 14.5% WER, Saint Gall dataset: 17.3% WER	Does not compare the proposed approach to existing fine-tuning methods for vision

		limited training data.				transformers.
2	TrOCR: Transformer-Based Optical Character Recognition with Pre-trained Models	The objective of this paper is to propose TrOCR, an end-to-end Transformer-based OCR model for text recognition with pre-trained CV and NLP models.	5,427 handwritten fonts to synthesize handwritten textline images by the TRDG2, an open-source text recognition data generator.	CRNN CNNbase d encoder and RNN-based decoder	The pre-trained models on the vision task improve the performance of text recognition models, and the pure Transformer models are better than the CRNN models and the Tesseract on this task	Lacks comparison with other Transformer-based OCR models.
3	Handwriting Transformers	Introduce a novel transformer-based approach, HWT, for styled handwritten text image generation. Aim to learn style-content entanglement, capture global and local style patterns through self-attention mechanisms, and achieve realistic results in handling arbitrary text lengths and diverse	-	Transformer-based network (HWT) with an encoder-decoder attention mechanism	HWT generates realistic styled handwritten text images	Limited comparison with existing methods for styled text generation.

		writing styles.				
4	AttentionHTR : handwritten text recognition based on attention encoder-decoder networks.	Develop an attention-based sequence-to-sequence model for handwritten word recognition, addressing data scarcity through transfer learning from pre-trained scene text images. Evaluate the proposed model on multi-writer datasets (IAM and Imgur5K) and conduct in-depth analyses of error cases.	IAM Dataset, Imgur5K Dataset	ResNet feature extraction and bidirectional LSTM-based sequence modeling stages forming an encoder.	Word Error Rate (WER): Achieved competitive results across different transfer learning strategies. IAM Dataset WER: 15.40% (word level) IAM Dataset CER: 6.50% (character level)	Does not investigate the impact of different pre-trained models on the performance.
5	Handwritten Text Recognition using Deep Learning	The research develops a system using CNNs and LSTMs to classify handwritten words, aiming to translate handwritten text into digital format for user-friendly document conversion.	IAM Handwriting Dataset, with 1500+ forms and 600+ writers, is the main resource, providing a diverse and verified dataset for model training.	Padding, rotation, VGG-19, RESNET-18/34 enhance word-level classification robustness with ADAM optimizer.	Word-level models (VGG-19, RESNET-18/34) succeed with large softmax challenges. Character-level classification enhances word recognition, promising efficient handwritten document conversion.	Does not explore the use of more recent and potentially more efficient architectures, such as Transformers.
6	Handwritten Text	Develop a Handwritten Text	IAM Handwriting	Utilized CNNs	Achieved over 90.3% accuracy,	Limited compari

	Recognition using Deep Learning with TensorFlow	Recognition (HTR) system for accurate word classification in digital conversion.	Dataset: 1500+ forms, 600+ writers, diverse and verified, with detailed transcription and coordinates.	(VGG-19, RESNET-18/34) for word-level classification, LSTMs for character segmentation, and Tesseract model for preprocessing.	successfully addressing challenges in large softmax layer output and offering a robust solution for handwritten text to digital conversion.	son with other HTR systems.
7	A Light Transformer-Based Architecture for Handwritten Text Recognition	Develop a lightweight Transformer-based architecture for Handwritten Text Recognition (HTR) with a focus on small datasets, aiming to achieve state-of-the-art results without the need for extensive training data.	The research employs the IAM Handwriting Dataset. With 400 text-line images samples training and evaluation.	Proposes Light Transformer architecture, integrating CNNs and Transformers. Employs hybrid loss (CTC + CE) for training.	Light Transformer achieves 5.70% CER on IAM test set without additional data. With synthetic data, CER improves to 4.76%, outperforming a larger variant (Large Transformer). Hybrid loss proves crucial for efficient training.	Does not provide a qualitative analysis of the generated outputs.
8	HTR-Flor: A	Develop a Gated-	The research	PyTorch	The Gated-CRNN	Limited

	Deep Learning System for Offline Handwritten Text Recognition	CRNN architecture to optimize Handwritten Text Recognition, achieving higher accuracy and efficiency with reduced parameters, particularly suitable for offline scenarios.	utilized the IAM Handwriting Database, a diverse collection of handwritten texts.	and TensorFlow for model development,. Gated-CRNN architecture, blending CNN and RNN	achieved state-of-the-art results on IAM dataset, outperforming existing methods. Notably, it demonstrated superior accuracy and reduced parameter count, making it suitable for resource-constrained environments.	explanation of the design choices for the Gated-CRNN architecture.
9	Recurrence-free unconstrained handwritten text recognition using gated fully convolutional network	The research aims to enhance unconstrained handwritten text recognition by introducing a Gated Fully Convolutional Network (GFCN), providing an efficient alternative to LSTM-based models, overcoming issues like excessive parameters and sequential processing during training and prediction.	Utilizing RIMES and IAM datasets, the research employs widely-accepted benchmarks for handwritten text recognition. Images are preprocessed by resizing to 64 pixels in height	Implemented in PyTorch, the GFCN model incorporates CTC loss, Depthwise Separable Convolutions (DSC), and various regularization techniques	The Gated Fully Convolutional Network (GFCN) excelled in unconstrained handwritten text recognition, achieving competitive CERs of 4.35% on RIMES and 7.99% on IAM datasets. With a low parameter count, GFCN displayed efficiency in training and	Lacks a detailed analysis of the learned features and filters in the GFCN, limiting the understanding of its decision-making process

			while retaining original width, forming the basis for training, validation, and testing.	. Training leverages the Adam optimizer on a GPU (V100 16GB), evaluating performance with CER and WER metrics.	prediction times, affirming its potential for text recognition tasks.	and identifying potential biases.
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2.8. Summary

Chapter II provides an in-depth exploration of advancements in Handwritten Text Recognition (HTR), driven by the integration of deep learning methodologies. The chapter introduces the evolution from Convolutional Neural Networks (CNNs) to Recurrent Neural Networks (RNNs) and the recent surge in transformer-based approaches. The narrative delves into challenges faced in recognizing diverse handwriting styles and highlights the transformative potential of large language models and transformers. The chapter discusses the problem background, emphasizing the limitations of traditional methods and the need for a new protagonist – transformer models. It outlines objectives such as a comprehensive literature review, adaptation of transformer architecture, diverse data curation, and model training. The storyline evolves with a focus on image processing, Optical Character Recognition (OCR), and the broader context of handwritten text recognition. The scientific overview emphasizes HTR's role in preserving cultural artifacts and healthcare applications. The chapter concludes with a review of related research, showcasing innovative approaches, challenges in historical document recognition, and the potential of transformer architectures in HTR.

Chapter Three

Methodology

3.CHAPTER III

3.1. Introduction

A sound empirical study of handwritten text recognition (HTR) requires more than the selection of a fashionable model: it requires an explicit account of the data, the preprocessing assumptions, the learning objectives, the evaluation criteria, and the comparisons against which claims will be judged. This chapter therefore sets out the methodological framework adopted for the present thesis. The research focuses on English line-level recognition using the IAM Handwriting Database and investigates transfer learning from a pretrained TrOCR transformer, whose Vision Transformer (ViT) encoder is paired with a BART text decoder. A convolutional recurrent network trained with Connectionist Temporal Classification (CRNN+CTC) provides a neural baseline of classical design, while Tesseract is retained as an optional classical optical character recognition reference when the runtime environment permits its use.

The chapter proceeds from dataset characterisation to preprocessing, model choice, metric definitions, and the planned experiment matrix. In keeping with conventional thesis organisation, no experimental scores are reported here; Chapter 4 is responsible for implementation detail, measured outcomes, and interpretive analysis.

3.2. Dataset

Reliable benchmarking in HTR depends on a corpus whose splits are stable and publicly documented. The study therefore employs the English IAM line corpus distributed as Hugging Face Teklia/IAM-line, which preserves the publisher’s official training, validation, and test partitions. Each sample consists of a line image together with its corresponding transcription. Local preparation yields organized image directories and label tables suitable for both training and evaluation, while a small demonstration subset is reserved for the interactive application described later in the thesis.

Property	Value
Language	English
Unit	Line images
Train / Val / Test	6,482 / 976 / 2,915
Modalities	PNG line image + transcription text
Source	Hugging Face Teklia/IAM-line

Table 3.1 — IAM line dataset (Teklia/IAM-line)

The test partition contains 2,915 lines. This full split constitutes the primary evaluation population of the thesis and is preferred over reduced samples that would weaken comparability with published IAM practice.



Figure 3.1 — Example IAM line

A representative English line image illustrating the visual domain addressed by the study.

3.3. Preprocessing

Preprocessing is kept intentionally conservative so that recognition performance can be attributed primarily to the model rather than to aggressive image engineering. Line images are converted to RGB and presented to the Hugging

Face TrOCR processor, which jointly handles visual normalisation and tokenisation for the encoder–decoder model. No custom binarisation pipeline is introduced. Mild geometric and photometric augmentation is applied exclusively during fine-tuning, on the premise that stochastic variation may aid adaptation while leaving the evaluation distribution unaltered. All reported test scores are therefore computed on unaugmented images.

3.4. Model selection

3.4.1. TrOCR as the primary model

TrOCR is adopted as the principal recognition architecture because it exemplifies the contemporary transfer-learning paradigm in HTR: an image transformer encodes local visual structure, while an autoregressive text transformer decodes character sequences under a language-model prior. The study begins from the publicly available checkpoint `microsoft/trocr-base-handwritten`, which has already been pretrained for English handwritten lines. Fine-tuning on the IAM training and validation partitions uses teacher-forced cross-entropy rather than CTC, with AdamW optimisation and early stopping guided by validation CER. This design choice is consequential: CTC remains associated only with the neural baseline, and is never treated as the training objective of TrOCR.

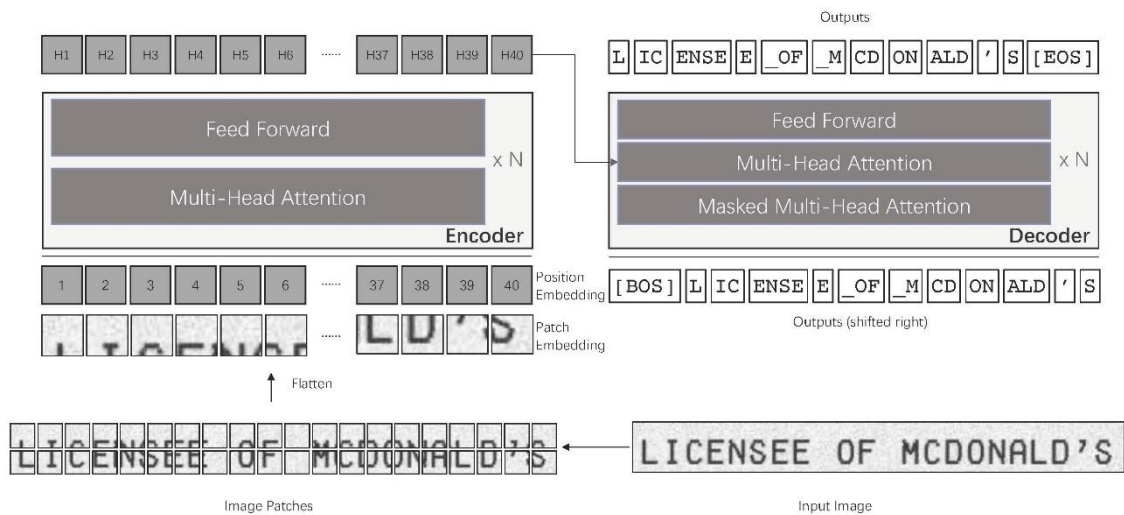


Figure 3.2 — TrOCR architecture (conceptual)

Image patches are encoded by a ViT encoder and decoded by a cross-attentive BART decoder into a token sequence.

3.4.2. CRNN+CTC as a neural baseline

To situate transformer performance against a familiar neural alternative, a CRNN with CTC is trained from scratch on the same IAM splits. The baseline is deliberately lightweight: convolutional feature extraction is followed by bidirectional recurrent modelling and CTC alignment. Its purpose is comparative clarity rather than architectural rivalry. A large performance gap, should it appear, would illuminate the benefit of large-scale pretrained transformers under matched data conditions.

3.4.3. Tesseract as an optional classical baseline

Tesseract, operated in line-oriented recognition mode, is included as a classical OCR reference. Because availability of the system binary can vary across machines, the methodology treats Tesseract as optional: when execution is impossible, Chapter 4 records a skipped status instead of fabricating scores. This rule preserves methodological honesty without forcing an incomplete software dependency to dictate the scientific narrative.

3.5. Experiment design and evaluation metrics

The empirical programme is organised as a small matrix of complementary runs. Primary scientific claims rest on full-test evaluation of pretrained TrOCR, fine-tuned TrOCR, and the CRNN+CTC baseline. The optional Tesseract run and the demonstration subset support completeness and application discussion, respectively, but do not redefine the primary evidence base.

ID	Run	Role	Evaluation sample
E1	Pretrained TrOCR	Primary transformer baseline	Full test (n=2,915)
E2	Fine-tuned TrOCR	Adaptation experiment	Full test (n=2,915)
E3	CRNN+CTC	Neural baseline	Full test (n=2,915)
E4	Tesseract	Optional classical baseline	Full test (n=2,915)
E5	Demonstration subset	Application support	Demo lines only

Table 3.2 — Designed experiment matrix

Two metrics dominate the evaluation. Character Error Rate (CER) quantifies the normalised edit distance at character level and remains the principal indicator of transcription fidelity in HTR. Word Error Rate (WER) provides a complementary word-level view of recognition difficulty. Both are computed after case-sensitive whitespace normalisation; transcriptions are not lowercased, so that the metric reflects orthographic detail rather than an artificially simplified string space. Corpus CER is defined as the ratio of total character edits to total reference characters across the evaluated set.

Engineering smoke tests with severely reduced sample limits may be used during development, but they are excluded from thesis claims. All numbers advanced as research evidence must originate from the full GPU protocol on the complete test partition.

3.6. Conclusion

This chapter has established the methodological foundation of the thesis. The research design centres on English line-level recognition with the IAM Handwriting Database, using the official training, validation, and test partitions so that evaluation rests on a complete and publicly documented population. Preprocessing remains deliberately light, preserving the primacy of the recognition model rather than of custom image engineering. TrOCR is adopted as the principal transfer-learning system, while CRNN+CTC supplies a controlled neural baseline and Tesseract is retained only as an optional classical reference. Character and word error rates, computed under a shared case-sensitive definition, constitute the criteria by which later claims will be judged.

Chapter Four

Implementation & Results

4. CHAPTER III

4.1. Purpose and contribution of the chapter

Where Chapter 3 defined the research design, the present chapter reports how that design was realised and what was observed. The contribution is best understood as an aligned research-and-application study rather than as the invention of a new HTR architecture. In concrete terms, the chapter offers four interlocking outcomes: a reproducible English IAM training and evaluation workflow; a fully logged TrOCR fine-tuning experiment on the official training and validation partitions; a controlled comparison against a CRNN+CTC baseline on the identical full test set of 2,915 lines; and an interactive demonstration that places the recogniser in an applied setting. Comparative reference to published IAM CER figures is included for contextual scale, without any claim of protocol-identical superiority.

4.2. Implementation overview

The implementation follows the methodological commitments of Chapter 3. Dataset handling loads line images and transcriptions for training and evaluation. Model utilities load either the public TrOCR checkpoint or a locally fine-tuned directory and generate transcriptions. A dedicated training routine adapts TrOCR with cross-entropy, AdamW, and early stopping on validation CER. A separate CRNN+CTC module supplies the neural baseline. Shared metric code computes case-sensitive CER and WER after whitespace normalisation. Evaluation runners execute the experiment matrix, while a Gradio interface exposes recognition for interactive demonstration. Detailed module listings are placed in the appendix so that the chapter body may remain focused on scientific substance rather than software documentation.

Component	Role in the study
Dataset I/O	Provides line images and transcriptions for train/eval
TrOCR loading and generation	Instantiates Hub or local checkpoints for inference
TrOCR fine-tuning	Adapts the pretrained model on IAM train/val
CRNN+CTC baseline	Trains and evaluates the neural reference system
Metrics	Computes CER and WER under a shared definition
Evaluation runners	Executes the designed experiment matrix
Demonstration interface	Supports interactive English line transcription

Table 4.1 — Principal implementation components

Technologies covered in detail

Technology	Key reason documented
Python 3.9+/3.10	ML ecosystem standard; required by all dependencies
PyTorch + torchvision	TrOCR/Hugging Face native runtime; CUDA support
Hugging Face Transformers	Official TrOCR implementation; avoids reimplementing ViT+BART
Hugging Face Hub	Model distribution without storing weights in Git
TrOCR (ViT + BART)	Thesis focus model; IAM-pretrained checkpoint
Gradio 4.x	ML demo UI without custom frontend development
Pillow	Standard image format for Gradio + TrOCRProcessor
jiwer	Dedicated CER/WER library; chosen over manual or evaluate
Hugging Face datasets	Official IAM splits from Teklia/IAM-line
Parquet + PyArrow	Efficient local caching between download and export
pandas	Tabular analysis of experiment outputs
PyYAML	Reproducible, human-readable experiment configs
Tesseract + pytesseract	Classical baseline on identical IAM data
matplotlib	Thesis CER comparison figures
setuptools / pyproject.toml	Editable install; explicit dependencies

JSON + CSV	Traceable, tool-agnostic experiment artifacts
functools.lru_cache	Avoid reloading model on every Gradio click

Table 4.2 summarizes the technology stack.

4.3. Experimental procedure

4.3.1. TrOCR fine-tuning

Fine-tuning began from microsoft/trocr-base-handwritten and used the full IAM training and validation sets (6,482 and 976 lines). Optimisation employed AdamW with a learning rate of 5×10^{-6} , a micro-batch size of 1, and gradient accumulation of 8, yielding an effective batch size of 8. A maximum of eight epochs was planned, with early-stopping patience of two validations. Training ran on CUDA and required approximately 88.7 hours of wall-clock time.

Under this schedule, validation CER reached its best value of 0.0137 after the first epoch and then deteriorated (0.0162 in epoch 2; 0.0187 in epoch 3). Early stopping therefore terminated training after three epochs, and the exported model corresponds to the best validation checkpoint rather than the final epoch weights. This behaviour is material to later interpretation: the adaptation run was completed and logged, yet it did not exhibit sustained validation improvement.

4.3.2. CRNN+CTC training

The CRNN baseline was trained from scratch on the same IAM partitions with learning rate 1×10^{-3} , batch size 8, and fixed input geometry of 32×128 . Eight epochs were executed under early-stopping patience, with a character set of size 79. Best validation CER remained high (approximately 0.80), already signalling limited competitiveness relative to pretrained TrOCR before test evaluation.

4.3.3. Evaluation matrix

Evaluation followed the design of Chapter 3. Measured full-test claims rest on pretrained TrOCR (E1), fine-tuned TrOCR (E2), CRNN+CTC (E3), and Tesseract (E4), each assessed on the identical IAM test split of 2,915 lines. The demonstration run (E5) supports the application section only: it is qualitative, may use a tiny demo subset, and is never used as a primary CER source for the thesis.

4.4. Results

Table 4.2 reports the primary measured outcomes.

Method	CER	WER	n
TrOCR pretrained	4.72%	11.64%	2915
TrOCR fine-tuned	4.96%	12.46%	2915
CRNN+CTC	81.82%	96.56%	2915
Tesseract	56.33%	91.35%	2915

Table 4.3 — Measured results on the full IAM English test set

The pretrained transformer attains a CER of 4.72% and a WER of 11.64%. After fine-tuning, CER and WER become 4.96% and 12.46%, respectively—slightly worse than the Hub checkpoint under the logged early-stopped schedule. The lightweight CRNN+CTC baseline yields a CER of 81.82% and a WER of 96.56%. Classical Tesseract, evaluated in line mode on the same split, yields a CER of 56.33% and a WER of 91.35%.

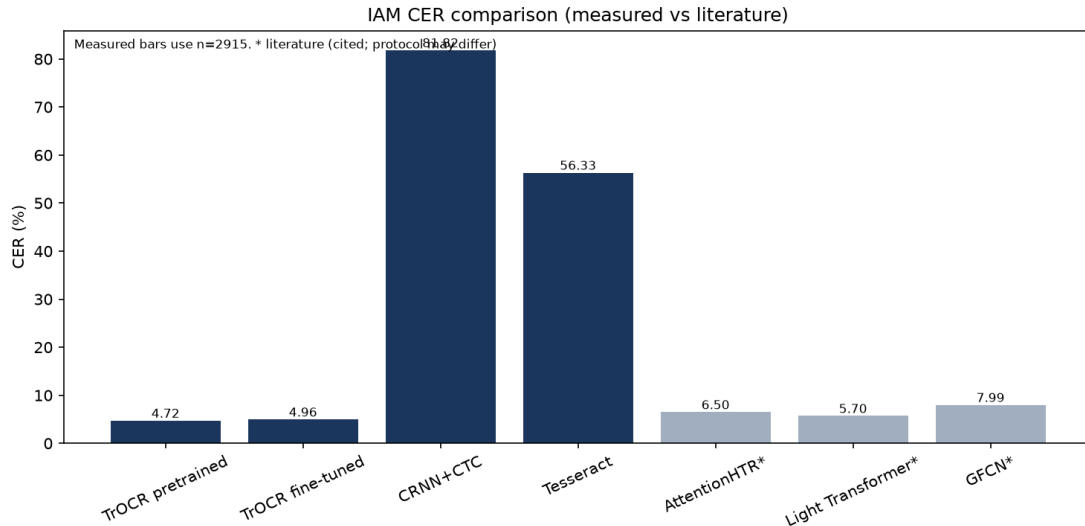


Figure 4.1 — CER comparison

Measured CER values for the systems evaluated in this thesis, with selected literature figures shown separately for contextual comparison.

Method	CER (IAM)	Source
AttentionHTR	6.50%	Kass & Vats, 2022
Light Transformer	5.70%	Barrère et al., 2022
GFCN	7.99%	Coquenot et al., 2020

Table 4.4 — Literature CER figures cited for context

These published values help the reader locate the present scores within the broader IAM literature. They are not re-implemented under the preprocessing and split conventions of this repository, and the thesis therefore does not assert a matched state-of-the-art ranking.

4.5. Discussion and analysis

4.5.1. Pretrained TrOCR as the principal empirical result

The pretrained TrOCR evaluation constitutes the central positive finding of the study. A CER of 4.72% on 2,915 previously unseen IAM lines indicates that a publicly released transformer HTR checkpoint can deliver strong line-level recognition when assessed under a complete and transparent local protocol. For a master’s investigation concerned with transfer learning rather than architectural invention, this result is scientifically

meaningful: it shows what current pretrained models already achieve before any student-led adaptation is attempted.

4.5.2. Interpretation of the fine-tuning outcome

Fine-tuning did not improve test recognition. The adapted model’s CER of 4.96% is slightly higher than that of the Hub checkpoint. Several considerations make this outcome intelligible. First, the starting checkpoint is already specialised for English handwriting, so residual headroom on IAM is limited. Second, validation CER worsened after the first epoch, and early stopping rightly preferred the earlier checkpoint; continued training under the logged schedule was not justified by the validation signal. Third, the optimisation used a small micro-batch with gradient accumulation, which may be adequate for a feasibility study yet suboptimal for extracting further gains. The appropriate academic reading is therefore that fine-tuning was executed and measured, and that its effect under these conditions was neutral to slightly adverse—not that adaptation is universally unhelpful, nor that the experiment failed to run.

4.5.3. The CRNN+CTC baseline in perspective

The CRNN+CTC system’s CER of 81.82% confirms a substantial gap relative to TrOCR under matched data and metrics. This outcome should be read carefully: the baseline is intentionally lightweight—a compact CNN–BiLSTM–CTC model trained from scratch with fixed 32×128 geometry—and is not presented as an optimised competitive neural HTR system. Its role is comparative anchoring. The large error rate therefore illustrates the advantage of large-scale pretrained transformers in this setting; it does not imply that every CTC architecture would fail equally.

4.5.4. Classical Tesseract as a reference baseline

Tesseract provides a classical OCR point of comparison on the same full test partition, reaching CER 56.33% and WER 91.35%. The gap to pretrained TrOCR is large, as expected when a general-purpose OCR engine is applied to unconstrained handwriting line images without handwriting-specific adaptation. The result strengthens the case for

transformer HTR while remaining modest in claim: Tesseract is a classical reference, not a tuned handwriting specialist under this protocol.

4.5.5. Error behaviour and qualitative observations

Inspection of high-CER predictions reveals recurring difficulties with very short references, punctuation-dominated or dash-like strings, and rare or hyphenated tokens. Such lines can inflate per-example CER even when corpus-level error remains comparatively low. These observations caution against over-interpreting isolated failures and reinforce the primacy of aggregate CER/WER on the full test partition.

4.5.6. Limitations affecting interpretation

Interpretation remains bounded by the study’s design. The work addresses English line-level IAM only. Literature comparisons are contextual rather than protocol-matched. The fine-tuning schedule was short and early-stopped, and should not be mistaken for an exhaustive hyperparameter search. The CRNN baseline was kept simple by design. Application/demo scores—especially any smoke or synthetic subset metrics—are illustrative only and are excluded from Table 4.2 and from primary thesis claims.

4.6. Application prototype

Beyond offline evaluation, the study includes a Gradio-based demonstration for interactive English line transcription, optionally with reference-based CER when ground truth is supplied. The prototype is an applied complement to the experimental programme: it shows that the recognition stack can be used by a non-specialist interface. Scientific claims remain dependent on the full-test E1–E4 runs reported above, not on demo-subset CER.

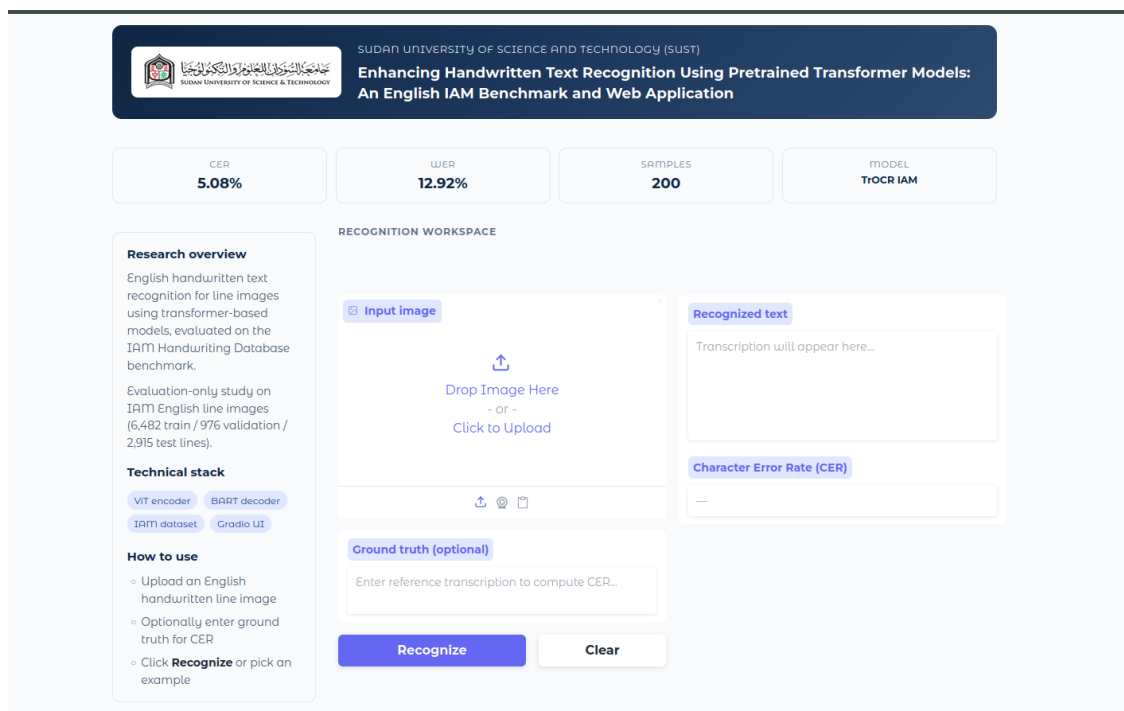


Figure 4.2 — Application Gradio UI

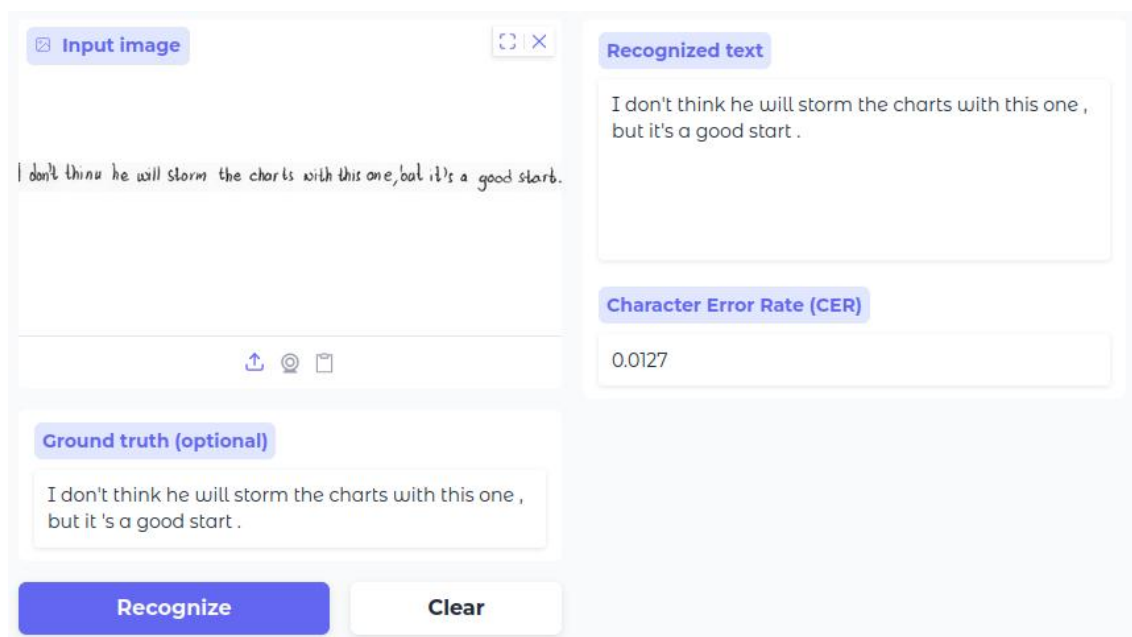


Figure 4.3 — Application Gradio UI - example

4.7. Reproducibility statement

All quantitative statements in this chapter are intended to be recoverable from the logged experiment artefacts of the project. Training hyperparameters are taken from the fine-tuning and CRNN training logs; test metrics are taken from the corresponding evaluation records. Reproduction follows the project’s documented GPU protocol and evaluation commands. In this way, Chapter 4 remains answerable to the methodological design of Chapter 3 and prepares the reflective synthesis developed in Chapter 5.